

# Degassing of organic solvents in Young-type flask

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**Tags:** AE Degassing  
**Category:** Protocol  
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## Note

This is for solvents which are not in the SPS! If the solvent is available in the SPS then use: [Protocol - Collecting of SPS solvents](#)

## Goal

Degassing of fluids using argon.

## Prerequisites and preparation

Solvent (500 mL)

Young-type flask (500 mL)

double tip metal canula

Molar Sieve (3 A)

## Steps

| Step number                            | Step description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Preparation                         | Equipp the Young type flask with a stopper and a septum. Dry the flask in the heat under vacuum three times.                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2. Molar Sieve                         | Add dried Molar sieve (approx. 1/5 of the total volume) to the flask. Set the flask under vacuum three times. Use a Filter frit to prevent dust going into the Schlenk line.                                                                                                                                                                                                                                                                                                                                             |
| 3. Prepare the transfer of the solvent | <p>Sekurate a needle, which is connected via a adapter to the schlenk line. For this seal the end with a gummi plug. Put the needle in the solvent container. Put a double tip metal cannula through the septum of the container and let argon flush through it for approx. 1 min.</p> <p>For small amounts the transfer is better done using syringes.</p>                                                                                                                                                              |
| 4. Transfer the solvent                | <p>Stick the cannula through the septum of the young type flask. Make sure that the cannula reaches into the flask. Stick another needle through the septum and reduce the argon flow of the young type flask, till the solvent gets transferred. Make sure, that still a tangible flow of argon leaves the needle.</p> <p>When the solvent has be transfered first open the argon of the young type flask completly, then remove the needle from the flask. Subsequently remove the cannula and close both vessels.</p> |

|                            |                                                                                                                                                                                                                                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5. Preparing the degassing | Sekurate a needle, which is connected via a adapter to the schlenk line (a long metal cannula is needed). Push it through the septum into the gas phase above the solvent. Stick another needle through the septum. Close the stopcock and push the needle attached to the Schlenk line into the solvent. |
| 6. Degassing               | Close the stop cock and push the needle atteched to the Schlenk line into the solvent. Let argon bubble through the solution. For non volitale solvents for 30 min. For volitale solvents the degassing time can be reduced to 20 to 25 min to avoid the loss of to much solvent.                         |
| 7. Finish                  | Remove the needle out of the liquid. Open the stop cock. Remove the Compensating cannula. Close the stop cock of the needle. Remove the needle attached to the Schlenk line. Close the flask.                                                                                                             |
| 8. Detemine water content  | (After approx. 1 week the water content is determined according to <a href="#">[Protocol] KFT (Karl-Fischer-Titration)</a> )                                                                                                                                                                              |

## Linked resources

Protocol - [KFT \(Karl-Fischer-Titration\)](#)

Protocol - [Collecting of SPS solvents](#)



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